

Single Factor Analysis

Analysis 1: Impact of Study Hours on Exam Performance

Objective

To analyze whether the number of hours studied has a significant impact on students' exam scores.

Hypothesis

Students who study for longer periods are expected to achieve higher exam scores.

Results

- Students were divided into three groups based on study hours: **Low (<10 hours)**, **Medium (10–19 hours)**, and **High (20+ hours)**.
- The **High** study group achieved the highest average exam score (**68.49**).
- The **Medium** study group achieved an average exam score of (**65.99**).
- The **Low** study group recorded the lowest average exam score (**63.82**).

Conclusion

The analysis indicates a positive relationship between study hours and exam performance. Students who dedicated more time to studying generally achieved higher exam scores, supporting the initial hypothesis.

	AZ Attendance_Level	123 Students	123 Average_Score	123 Min_Score	123 Max_Score
1	Poor	1,573	64.21	55	95
2	Good	3,288	67.14	58	100
3	Excellent	1,745	70.12	62	99

Analysis 2: Impact of Attendance on Exam Performance

Objective

To evaluate the relationship between students' attendance levels and their exam performance.

Hypothesis

Students with higher attendance rates are expected to achieve better exam scores than those with lower attendance.

Results

Students were classified into three attendance categories: **Poor** (less than 70%), **Good** (70%–89%), and **Excellent** (90% and above).

The analysis showed that students with **Poor** attendance had an average exam score of **64.21**. Students with **Good** attendance achieved an average exam score of **67.14**, while students with **Excellent** attendance recorded the highest average exam score of **70.12**.

The results demonstrate a gradual increase in average exam scores as attendance levels improve.

Conclusion

The analysis indicates a positive relationship between attendance and academic performance. Students who attended classes more consistently achieved higher average exam scores, supporting the hypothesis that regular attendance contributes to better academic outcomes.

	AZ Attendance_Level	123 Students	123 Average_Score	123 Min_Score	123 Max_Score
1	Poor	1,573	64.21	55	95
2	Good	3,288	67.14	58	100
3	Excellent	1,745	70.12	62	99

Analysis 3: Impact of Previous Academic Performance on Current Exam Results

Objective

To investigate whether students' previous academic performance influences their current exam results and to determine how many students showed improvement or decline compared to their previous scores.

Hypothesis

Students with higher previous academic scores are expected to achieve better results in the current exam. However, some students may significantly improve or decline due to other factors such as study habits, attendance, motivation, and access to learning resources.

Results

Students were first categorized into three groups based on their previous academic performance: **Low**, **Medium**, and **High**. The average current exam scores were **66.39**, **66.93**, and **67.91**, respectively. Although students with higher previous scores achieved slightly better average exam scores, the difference between the three groups was relatively small, indicating that previous academic performance alone is not a strong predictor of the current exam result.

To gain deeper insight, student performance was compared directly with their previous scores. The analysis showed that **4,332 students (approximately 65.6%)** scored lower than their previous performance, with an average decline of **16.24** points. In contrast, **1,257 students (approximately 19.0%)** improved their scores by an average of **4.99** points, while **882 students (approximately 13.4%)** achieved a **significant improvement** of at least 10 points, with an average increase of **13.97** points. Additionally, **135 students (approximately 2.0%)** maintained the same score.

Conclusion

The results suggest that previous academic performance has only a limited effect on current exam outcomes. While students with higher previous scores obtained slightly higher average exam scores, a considerable number of students experienced substantial changes in performance. The presence of

many students with significant improvement and others with notable declines indicates that additional factors—such as study hours, attendance, motivation, teacher quality, and access to educational resources—likely play a more influential role in determining current academic performance than previous scores alone.

	AZ Previous_Performance	123 Number_of_Students	123 Average_Exam_Score	123 Min_Score	123 Max_Score
1	Low	1,235	66.39	58	95
2	Medium	2,657	66.93	55	100
3	High	2,714	67.91	58	99

	AZ Performance_Change	123 Students	123 Average_Change
1	Declined	4,332	-16.24
2	Improved	1,257	4.99
3	Significant Improvement	882	13.97
4	No Change	135	0

Analysis 4: Impact of Sleep Hours on Student Performance Improvement

Objective

To investigate whether the number of hours students sleep influences the change in their academic performance compared to their previous scores.

Hypothesis

Students who obtain the recommended amount of sleep (6–8 hours) are expected to achieve better academic performance than those who sleep for shorter or longer durations.

Results

Students were classified into three sleep categories: **Insufficient Sleep** (less than 6 hours), **Recommended Sleep** (6–8 hours), and **Extended Sleep** (more than 8 hours).

The analysis showed that the average exam scores were remarkably similar across all three groups. Students with **Insufficient Sleep** achieved the highest average exam score (**67.40**), followed by those with **Recommended Sleep** (**67.21**), while students with **Extended Sleep** recorded an average score of **67.15**. The maximum difference between the groups was only **0.25 points**, indicating minimal variation in academic performance.

When comparing current exam scores with previous academic performance, students with **Extended Sleep** experienced the smallest average decline (**-7.21** points), whereas students in the **Recommended Sleep** and **Insufficient Sleep** groups showed similar declines of **-7.95** and **-7.99** points, respectively.

Conclusion

The analysis does not provide evidence of a strong relationship between sleep duration and academic performance in this dataset. Despite the common expectation that sleeping 6–8 hours lead to better academic outcomes, the observed differences in exam scores and performance changes were minimal. These findings suggest that sleep duration alone is not a significant predictor of academic success and that other factors, such as study hours, attendance, parental involvement, teacher quality, and access to educational resources, have a greater influence on students' performance.

	A2 Sleep_Category ▾	123 Number_of_Students ▾	123 Average_Exam_Score ▾	123 Average_Improvement ▾	123 Min_Score ▾	123 Max_Score ▾
1	Insufficient Sleep (<6 Hours)	1,004	67.4	-7.99	58	100
2	Recommended Sleep (6-8 Hours)	4,515	67.21	-7.95	55	99
3	Extended Sleep (>8 Hours)	1,087	67.15	-7.21	57	98

Analysis 5: Impact of Physical Activity on Exam Performance

Objective

To investigate whether the level of physical activity influences students' academic performance and changes in their exam scores.

Hypothesis

Students who engage in regular physical activity are expected to achieve better academic performance due to the positive effects of exercise on concentration, mental health, and overall well-being.

Results

Students were grouped according to the number of hours of physical activity, ranging from **0 to 6 hours**. The majority of students participated in **2 to 4 hours** of physical activity, while only **32 students** reported **6 hours**, making this group too small for reliable conclusions.

The average exam scores remained relatively stable across all activity levels, ranging from **66.97** to **68.09**. Similarly, the average change in performance compared with previous scores ranged from **-12.80** to **-6.65** points.

Students with **5 hours** of physical activity recorded the smallest average decline in performance (**-6.65**), while students with **0 hours** experienced the largest average decline (**-12.80**). However, because the differences among the larger groups (2–4 hours) were minimal, the results do not indicate a strong relationship between physical activity and exam performance.

Conclusion

The analysis suggests that physical activity has only a limited impact on academic performance in this dataset. Although students who engaged in moderate levels of physical activity showed slightly better outcomes, the differences were relatively small. Additionally, categories with very few students, such as the 6-hour group, should not be used to draw general conclusions. Overall, factors such as study hours, attendance, motivation, and access to educational resources appear to have a stronger influence on students' exam performance than physical activity alone.

	123 Physical_Activity ▾	123 Number_of_Students ▾	123 Average_Exam_Score ▾	123 Average_Improvement ▾	123 Min_Score ▾	123 Max_Score ▾
1	0	46	67.52	-12.8	58	80
2	1	421	66.97	-7.87	57	97
3	2	1,627	67.19	-7.73	56	100
4	3	2,544	67.18	-8.14	55	99
5	4	1,575	67.31	-7.58	57	98
6	5	361	67.6	-6.65	59	95
7	6	32	68.09	-7.44	61	75

Analysis 6: Impact of Tutoring Sessions on Exam Performance

Objective

To examine whether attending tutoring sessions contributes to improving students' academic performance.

Hypothesis

Students who attend more tutoring sessions are expected to achieve higher exam scores and demonstrate better academic improvement.

Results

Students were grouped according to the number of tutoring sessions they attended, ranging from **0 to 8 sessions**. The analysis showed a gradual increase in average exam scores as the number of tutoring sessions increased from **0 to 4 sessions**. Students with no tutoring sessions achieved an average exam score of **66.49**, while students attending **4 sessions** achieved an average score of **68.23**.

Similarly, the average decline from previous scores became smaller as tutoring sessions increased, improving from **-9.18** for students with no tutoring sessions to **-6.01** for students attending four sessions. This suggests that regular tutoring may help students maintain or improve their academic performance.

Although students attending **5 to 8 sessions** achieved relatively high average exam scores, these groups contained very few students (102, 18, 7, and 1 student, respectively). Therefore, these results should be interpreted with caution and should not be generalized to the entire population.

Conclusion

The analysis indicates a positive relationship between tutoring sessions and academic performance, particularly for students attending up to four tutoring sessions. Students who participated in regular tutoring generally achieved higher average exam scores and experienced smaller declines in performance compared with students who did not receive tutoring. However, conclusions regarding students with more than four tutoring sessions are less reliable due to the limited number of observations in those groups.

	123 Tutoring_Sessions ▼	123 Number_of_Students ▼	123 Average_Exam_Score ▼	123 Average_Improvement ▼	123 Min_Score ▼	123 Max_Score ▼
1	0	1,513	66.49	-9.18	56	98
2	1	2,179	66.98	-7.84	55	99
3	2	1,649	67.57	-7.72	58	98
4	3	836	67.89	-6.35	58	100
5	4	301	68.23	-6.01	59	99
6	5	102	68.75	-6.66	60	77
7	6	18	71.67	-10.94	64	78
8	7	7	69.86	-10.86	63	75
9	8	1	69	-9	69	69

Analysis 7: Impact of Motivation Level on Exam Performance

Objective

To investigate the relationship between students' motivation levels and their academic performance.

Hypothesis

Students with higher motivation levels are expected to achieve better exam scores and demonstrate better academic performance than students with lower motivation levels.

Results

Students were classified into three motivation levels: **High**, **Medium**, and **Low**. The analysis showed that students with **High** motivation achieved the highest average exam score (**67.70**), followed by students with **Medium** motivation (**67.33**), while students with **Low** motivation recorded the lowest average exam score (**66.73**).

When comparing current exam scores with previous academic performance, students with **Medium** motivation experienced the smallest average decline (**-7.46** points), followed by students with **High** motivation (**-7.88** points). Students with **Low** motivation showed the largest average decline (**-8.46** points), indicating that lower motivation may be associated with a greater decrease in academic performance.

Although the differences in average exam scores between the three groups were relatively small, the overall trend consistently shows that students with higher motivation tend to perform slightly better academically than those with lower motivation.

Conclusion

The analysis suggests a positive relationship between motivation and academic performance. Students with higher motivation generally achieved better exam scores and experienced smaller declines in performance compared with less motivated students. However, the differences between the groups were

modest, indicating that motivation is an important contributing factor but should be considered alongside other variables such as study hours, attendance, teacher quality, and access to educational resources when evaluating student performance.

	AZ motivation_level	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score
1	High	1,319	67.7	-7.88	57	98
2	Medium	3,351	67.33	-7.46	55	100
3	Low	1,936	66.73	-8.46	57	99

Analysis 8: Impact of Family Income on Exam Performance

Objective

To investigate whether family income influences students' academic performance and their improvement compared to previous academic results.

Hypothesis

Students from higher-income families are expected to achieve better academic performance due to greater access to educational resources, learning opportunities, and academic support.

Results

Students were classified into three family income levels: **High**, **Medium**, and **Low**. The analysis showed that students from **High-income** families achieved the highest average exam score (**67.82**), followed by students from **Medium-income** families (**67.33**), while students from **Low-income** families recorded the lowest average exam score (**66.85**).

When comparing current exam scores with previous academic performance, students from **High-income** families experienced the smallest average decline (**-6.77** points). Students from **Medium-income** families showed an average decline of **-7.82** points, while students from **Low-income** families recorded the largest average decline (**-8.36** points).

The number of students in the three income groups was relatively balanced, making the comparison more reliable than analyses involving very small sample sizes.

Conclusion

The analysis indicates a moderate positive relationship between family income and academic performance. Students from higher-income families generally achieved slightly higher exam scores and experienced smaller declines in performance compared with students from medium- and low-income families. Although the differences were not substantial, the consistent trend suggests that family income may contribute to academic success, likely through improved access to educational resources and learning support. However, family income should be considered alongside other factors such as study

hours, attendance, motivation, and teacher quality when evaluating overall student performance.

	AZ income_level	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score
1	High	1,268	67.82	-6.77	58	100
2	Medium	2,666	67.33	-7.82	57	99
3	Low	2,672	66.85	-8.36	55	97

Analysis 9: Impact of Teacher Quality on Exam Performance

Objective

To investigate whether teacher quality influences students' academic performance and their ability to maintain or improve their results compared with previous academic performance.

Hypothesis

Students taught by higher-quality teachers are expected to achieve better exam scores and experience smaller declines in academic performance than students taught by lower-quality teachers.

Results

Students were grouped into three teacher quality levels: **High**, **Medium**, and **Low**. The analysis showed that students taught by **High-quality** teachers achieved the highest average exam score (**67.66**), followed by those taught by **Medium-quality** teachers (**67.10**). Students with **Low-quality** teachers recorded the lowest average exam score (**66.75**).

When comparing current exam scores with previous academic performance, students in the **High** and **Medium** teacher quality groups experienced similar average declines (**-7.66** and **-7.67** points, respectively). In contrast, students taught by **Low-quality** teachers experienced a noticeably larger average decline of **-9.39** points.

The dataset also contained enough students in each category (1,946 in the High group, 4,003 in the Medium group, and 657 in the Low group), making the comparison reasonably reliable.

Conclusion

The analysis suggests a positive relationship between teacher quality and academic performance. Students taught by high-quality teachers generally achieved the highest average exam scores and maintained their academic performance better than students taught by low-quality teachers. Although the difference in average exam scores between the High and Medium groups was relatively small, students in the Low teacher quality group experienced a considerably larger decline in performance. This indicates that teacher quality

may play an important role in supporting students' academic achievement, particularly in helping them maintain consistent performance over time.

	AZ quality_level	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score
1	High	1,946	67.66	-7.66	58	98
2	Medium	4,003	67.1	-7.67	55	100
3	Low	657	66.75	-9.39	58	94

Analysis 10: Impact of Access to Educational Resources on Exam Performance

Objective

To examine whether access to educational resources influences students' academic performance and their ability to maintain or improve their performance compared with previous academic results.

Hypothesis

Students with greater access to educational resources are expected to achieve higher exam scores and experience smaller declines in academic performance than students with limited access to resources.

Results

Students were classified into three levels of access to educational resources: **High**, **Medium**, and **Low**. The analysis showed a clear and consistent trend across all three groups.

Students with **High** access to educational resources achieved the highest average exam score (**68.09**), followed by students with **Medium** access (**67.12**), while students with **Low** access recorded the lowest average exam score (**66.20**). The difference of nearly **1.9 points** between the High and Low groups is one of the largest observed among the analyzed factors.

When comparing current exam scores with previous academic performance, students with **High** access experienced the smallest average decline (**-7.29** points). In contrast, students with **Medium** and **Low** access showed larger average declines of **-8.05** and **-8.12** points, respectively.

Conclusion

The analysis indicates a strong positive relationship between access to educational resources and academic performance. Students with greater access to learning resources consistently achieved higher average exam scores and experienced smaller declines in performance compared with students who had limited access. These findings suggest that access to educational resources is one of the more influential factors affecting student achievement in this dataset and should be considered alongside study hours, teacher quality, attendance, and family income when evaluating academic performance.

	 AZ resource_level	 Number_of_Students	 Average_Exam_Score	 Average_Improvement	 Min_Score	 Max_Score
1	High	1,975	68.09	-7.29	56	99
2	Medium	3,318	67.12	-8.05	58	100
3	Low	1,313	66.2	-8.12	55	98

Analysis 11: Impact of Distance from Home on Exam Performance

Objective

To investigate whether the distance between a student's home and school affects academic performance.

Hypothesis

Students who live closer to school are expected to achieve better academic performance due to reduced commuting time, lower transportation-related stress, and improved attendance.

Results

Students were classified into three distance categories: **Near**, **Moderate**, and **Far**. The analysis showed that students living **Near** the school achieved the highest average exam score (**67.49**), followed by students living at a **Moderate** distance (**66.96**). Students living **Far** from school recorded the lowest average exam score (**66.46**).

When comparing current exam scores with previous academic performance, students living **Near** the school experienced the smallest average decline (**-7.68** points). Students in the **Moderate** group showed a slightly larger decline (**-7.89** points), while students living **Far** from school experienced the largest average decline (**-8.63** points).

The distribution of students across the categories was also sufficient for comparison, with **3,951** students in the Near group, **1,997** in the Moderate group, and **658** in the Far group.

Conclusion

The analysis indicates a moderate relationship between the distance from home and academic performance. Students living closer to school generally achieved slightly higher exam scores and maintained their academic performance better than students who lived farther away. Although the differences in average exam scores were relatively modest, the consistent increase in performance decline as distance increased suggests that commuting distance may have a negative impact on students' academic outcomes. However, this factor should be considered alongside other

influences such as study hours, attendance, teacher quality, and access to educational resources.

	AZ distance_level	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score
1	Near	3,951	67.49	-7.68	55	100
2	Moderate	1,997	66.96	-7.89	57	98
3	Far	658	66.46	-8.63	56	99

Analysis 12: Impact of Gender on Exam Performance

Objective

To examine whether gender has a significant influence on students' academic performance.

Hypothesis

There may be differences in academic performance between male and female students due to various social, behavioral, or educational factors.

Results

The analysis compared the academic performance of **Female** and **Male** students. Both groups achieved an identical average exam score of **67.23**, indicating no measurable difference in overall academic achievement.

Similarly, the average change in performance compared with previous scores was nearly identical. Female students experienced an average decline of **-7.81** points, while male students showed a slightly larger decline of **-7.85** points. The difference of only **0.04** points is negligible and does not indicate any meaningful variation between the two groups.

The dataset included **2,792 female** students and **3,814 male** students, providing a sufficiently large sample for a reliable comparison.

Conclusion

The analysis indicates that gender has virtually **no impact** on academic performance in this dataset. Both male and female students achieved the same average exam score and exhibited almost identical changes in performance compared with their previous academic results. These findings suggest that gender is not a significant factor influencing students' exam performance, while other factors such as study hours, attendance, teacher quality, motivation, and access to educational resources appear to play a much greater role.

	AZ gender_name	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score
1	Female	2,792	67.23	-7.81	57	100
2	Male	3,814	67.23	-7.85	55	99

Analysis 13: Impact of School Type on Exam Performance

Objective

To investigate whether the type of school (Public or Private) influences students' academic performance.

Hypothesis

Students attending private schools are expected to achieve slightly better academic performance due to potentially greater access to educational resources, smaller class sizes, and additional academic support.

Results

The analysis compared students enrolled in **Private** and **Public** schools. Students attending **Private** schools achieved a slightly higher average exam score (**67.29**) compared with students in **Public** schools (**67.21**). However, the difference in average exam scores was only **0.08 points**, indicating that school type had very little effect on overall academic achievement.

When comparing current exam scores with previous academic performance, students in **Private** schools experienced a smaller average decline (**-7.49** points) than students in **Public** schools (**-7.99** points). Although the difference is modest, it suggests that students in private schools maintained their academic performance slightly better over time.

The comparison is based on a large sample of **2,009** students from private schools and **4,597** students from public schools, making the results reliable.

Conclusion

The analysis suggests that school type has only a **limited influence** on academic performance. While students attending private schools achieved marginally higher average exam scores and experienced a slightly smaller decline in performance, the differences between the two groups were minimal. These findings indicate that the type of school alone is not a major determinant of academic success, and that other factors such as study hours, attendance, teacher quality, motivation, and access to educational resources have a greater impact on students' exam performance.

	AZ school_type_name	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score
1	Private	2,009	67.29	-7.49	56	100
2	Public	4,597	67.21	-7.99	55	99

Analysis 14: Impact of Internet Access on Exam Performance

Objective

To investigate whether having internet access influences students' academic performance.

Hypothesis

Students with internet access are expected to achieve better academic performance due to easier access to online learning resources, educational materials, and digital learning platforms.

Results

The analysis compared students with and without internet access. Students who had **Internet Access** achieved a higher average exam score (**67.29**) compared with students who did not have internet access (**66.47**), representing a difference of **0.82 points**.

When comparing current exam scores with previous academic performance, students with internet access experienced a smaller average decline (**-7.80** points) than students without internet access (**-8.35** points). This suggests that internet access may contribute to helping students maintain their academic performance.

The dataset included **6,108** students with internet access and **498** students without internet access. Although the "No Internet Access" group is considerably smaller, it still provides a reasonable basis for comparison.

Conclusion

The analysis indicates a positive relationship between internet access and academic performance. Students with internet access achieved slightly higher average exam scores and experienced a smaller decline in performance compared with students who lacked internet access. While the differences are not substantial, they consistently support the hypothesis that access to online educational resources can contribute to improved academic outcomes. However, internet access should be considered alongside other influential factors such as study hours, motivation, teacher quality, and access to educational resources.

	A2 access_name	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score	
1	Yes	6,108	67.29	-7.8	55	100	
2	No	498	66.47	-8.35	58	93	

Analysis 15: Impact of Extracurricular Activities on Exam Performance

Objective

To investigate whether participation in extracurricular activities influences students' academic performance.

Hypothesis

Students who participate in extracurricular activities are expected to achieve better academic performance due to the development of time management, teamwork, discipline, and social skills.

Results

The analysis compared students who participated in extracurricular activities with those who did not. Students who participated in extracurricular activities achieved a higher average exam score (**67.43**) compared with students who did not participate (**66.93**), representing a difference of **0.50 points**.

When comparing current exam scores with previous academic performance, students involved in extracurricular activities experienced a smaller average decline (**-7.68** points) than students who did not participate (**-8.06** points). This indicates that participation in extracurricular activities may contribute to maintaining students' academic performance.

The comparison is based on a large sample of **3,937** students who participated in extracurricular activities and **2,669** students who did not, making the results reliable.

Conclusion

The analysis suggests a modest positive relationship between participation in extracurricular activities and academic performance. Students who engaged in extracurricular activities achieved slightly higher average exam scores and experienced a smaller decline in performance than those who did not participate. Although the differences are relatively small, the consistent trend supports the hypothesis that involvement in extracurricular activities can have a positive contribution to students' overall academic achievement when combined with other factors such as study habits, attendance, and motivation.

	AZ activity_name	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score	
1	Yes	3,937	67.43	-7.68	57	100	
2	No	2,669	66.93	-8.06	55	97	

Analysis 16: Impact of Learning Disabilities on Exam Performance

Objective

To investigate whether learning disabilities influence students' academic performance and their ability to maintain or improve their performance compared with previous academic results.

Hypothesis

Students with learning disabilities may face additional academic challenges, which could result in lower exam scores and greater declines in performance if appropriate educational support is not available.

Results

The analysis compared students with and without learning disabilities. Students **without learning disabilities** achieved a higher average exam score (**67.34**) than students **with learning disabilities**, who recorded an average score of **66.27**. This represents a difference of approximately **1.07 points**.

When comparing current exam scores with previous academic performance, students without learning disabilities experienced an average decline of **-7.69** points, whereas students with learning disabilities showed a noticeably larger average decline of **-9.10** points. This suggests that students with learning disabilities faced greater challenges in maintaining their academic performance.

The comparison was based on **5,911** students without learning disabilities and **695** students with learning disabilities, providing a sufficiently large sample to support the analysis.

Conclusion

The analysis indicates that learning disabilities are associated with lower academic performance and a greater decline in exam scores compared with previous performance. While the difference in average exam scores is moderate, the larger decrease in performance among students with learning disabilities suggests that they may require additional academic support and appropriate learning accommodations to achieve outcomes comparable to

their peers. These findings highlight the importance of providing targeted educational assistance to support students with diverse learning needs.

	A2 disability_name	I23 Number_of_Students	I23 Average_Exam_Score	I23 Average_Improvement	I23 Min_Score	I23 Max_Score
1	No	5,911	67.34	-7.69	55	100
2	Yes	695	66.27	-9.1	57	89

Analysis 17: Impact of Parental Education Level on Exam Performance

Objective

To investigate whether parents' education level influences students' academic performance and their ability to maintain or improve their performance compared with previous academic results.

Hypothesis

Students whose parents have higher levels of education are expected to achieve better academic performance due to increased academic guidance, educational support, and a stronger learning environment at home.

Results

Students were grouped according to their parents' highest education level: **High School**, **College**, and **Postgraduate**. The analysis revealed a clear upward trend in academic performance as parental education level increased.

Students whose parents had a **Postgraduate** education achieved the highest average exam score (**67.97**), followed by students whose parents completed **College** (**67.32**). Students whose parents had only a **High School** education recorded the lowest average exam score (**66.89**).

A similar pattern was observed when comparing current exam scores with previous academic performance. Students with **Postgraduate-educated** parents experienced the smallest average decline (**-7.11** points), followed by students whose parents completed **College** (**-7.33** points). In contrast, students whose parents had a **High School** education experienced the largest average decline (**-8.43** points).

Conclusion

The analysis indicates a positive relationship between parental education level and students' academic performance. Students whose parents attained higher levels of education generally achieved higher exam scores and maintained their academic performance more effectively than students whose parents had lower educational attainment. These findings suggest that parental education may contribute to creating a supportive learning environment and should be considered as one of the important factors influencing academic success,

alongside study habits, motivation, teacher quality, and access to educational resources.

	A2 education_level	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score
1	Postgraduate	1,305	67.97	-7.11	57	98
2	College	1,989	67.32	-7.33	56	100
3	High School	3,312	66.89	-8.43	55	99

Analysis 18: Impact of Peer Influence on Exam Performance

Objective

To investigate whether peer influence affects students' academic performance and their ability to maintain or improve their performance compared with previous academic results.

Hypothesis

Students surrounded by positive peer influence are expected to achieve better academic performance than those influenced by neutral or negative peer groups, as supportive peers can encourage effective study habits, motivation, and academic engagement.

Results

Students were classified into three peer influence categories: **Positive**, **Neutral**, and **Negative**. The analysis revealed a consistent relationship between peer influence and academic performance.

Students with **Positive** peer influence achieved the highest average exam score (**67.61**), followed by students with **Neutral** peer influence (**67.20**). Students with **Negative** peer influence recorded the lowest average exam score (**66.56**), representing a difference of approximately **1.05 points** between the Positive and Negative groups.

A similar trend was observed when comparing current exam scores with previous academic performance. Students with **Positive** peer influence experienced the smallest average decline (**-7.05 points**), while students with **Neutral** peer influence showed a larger decline (**-8.19 points**). Students with **Negative** peer influence experienced the greatest decline (**-8.67 points**), indicating that unfavorable peer environments may negatively affect students' ability to maintain their academic performance.

Conclusion

The analysis indicates a clear positive relationship between peer influence and academic performance. Students surrounded by positive peers generally achieved higher average exam scores and experienced smaller declines in performance than students exposed to neutral or negative peer influence.

Among the social factors analyzed in this project, peer influence appears to be one of the more influential variables, suggesting that a supportive academic and social environment can contribute meaningfully to student success.

	A2 influence_name	I23 Number_of_Students	I23 Average_Exam_Score	I23 Average_Improvement	I23 Min_Score	I23 Max_Score
1	Positive	2,637	67.61	-7.05	57	100
2	Neutral	2,592	67.2	-8.19	57	97
3	Negative	1,377	66.56	-8.67	55	99

Analysis 19: Impact of Parental Involvement on Exam Performance

Objective

To investigate whether the level of parental involvement influences students' academic performance and their ability to maintain or improve their academic achievement.

Hypothesis

Students whose parents are more actively involved in their education are expected to achieve higher exam scores and experience smaller declines in academic performance than students with lower levels of parental involvement.

Results

Students with **High** parental involvement achieved the highest average exam score (**68.09**), followed by students with **Medium** parental involvement (**67.10**). Students with **Low** parental involvement recorded the lowest average exam score (**66.33**), resulting in a difference of approximately **1.76 points** between the High and Low groups.

A similar trend was observed when comparing current exam scores with previous academic performance. Students with **High** parental involvement experienced the smallest average decline (**-6.66** points), while students with **Medium** parental involvement showed a larger decline (**-7.92** points). Students with **Low** parental involvement experienced the greatest decline (**-9.32** points), indicating that limited parental involvement may negatively affect students' ability to maintain their academic performance.

Conclusion

The analysis indicates that parental involvement is one of the strongest factors associated with academic performance in this dataset. Students with higher levels of parental involvement consistently achieved higher average exam scores and experienced significantly smaller declines in performance compared with students receiving less parental support. These findings suggest that active parental engagement plays an important role in supporting students' academic success and should be considered alongside other key factors such as study hours, teacher quality, motivation, and access to educational resources.

	A2 involvement_name	123 Number_of_Students	123 Average_Exam_Score	123 Average_Improvement	123 Min_Score	123 Max_Score
1	High	1,908	68.09	-6.66	57	100
2	Medium	3,362	67.1	-7.92	55	97
3	Low	1,336	66.33	-9.32	56	98

Influence ranking

Strong Factors

1. Study Hours
2. Attendance
3. Parental Involvement
4. Access to Educational Resources
5. Parental Education Level

Moderate Factors

6. Teacher Quality
7. Family Income
8. Peer Influence
9. Tutoring Sessions
10. Motivation Level
11. Internet Access
12. Extracurricular Activities
13. Learning Disabilities

Weak Factors

14. Distance from Home
15. Physical Activity
16. Sleep Hours
17. School Type
18. Gender
19. Previous Academic Performance

The influence ranking is based on descriptive statistical analysis of average exam scores, performance changes, and consistency of trends across different categories. It provides an indication of the relative importance of each factor within this dataset and should not be interpreted as proof of causation. More advanced statistical methods, such as correlation analysis or regression modeling, would be required to quantify the exact impact of each factor.

